KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

YEAR 7 MATHEMATICS · CHAPTER 11 PROBABILITY

Complementary events

EXTENSION WORKSHEET · WORK ON YOUR OWN

MA4-PRO-C-01
Solves problems involving the probabilities of simple chance experiments

KEY WORDS

complement	all the outcomes that are **not** in the event.
more than 4	5, 6, … (4 is **not** included). Its complement is "4 or less".
at least 4	4 or more (4 **is** included). Its complement is "less than 4".
prime	has exactly two factors: 2, 3, 5, 7, … (1 is not prime).

How to use this sheet: read each grey **Example**, then do the blue **Round** under it. The first question in a round is started for you.

Notes: If the only outcomes are A, B and C, then P(A) + P(B) + P(C) = 1. Probabilities are never below 0 or above 1.

FORMULAS

P(not A) = 1 − P(A)

P(A) + P(not A) = 1

All the outcomes = 1 = 100%



35% + 65% = 100%

Tip: $1 = \frac{8}{8} = 1.00 = 100\%$

EXAMPLE 1 — Complements with "more than" and "at least"

Use me for **Round 1** ↓

Cards numbered 1 to 10. **Event:** picking a number greater than 6. Write the complement and list its outcomes.

1. List all the outcomes and cross out the event (7, 8, 9, 10): 1 2 3 4 5 6 ~~7~~ ~~8~~ ~~9~~ ~~10~~
2. **Complement: 6 or less. Outcomes: 1, 2, 3, 4, 5, 6** — Careful: 6 is in the complement. It is not "less than 6".

ROUND 1 — Medium — Extension

↑ Use Example 1

	What happens	Complement (in words)	Outcomes in the complement
1	A die is rolled. **Event:** rolling more than 2	rolling ____ or less	____, ____
2	A die is rolled. **Event:** rolling a prime number	_____	_____
3	This spinner is spun. **Event:** a multiple of 3 	_____	_____
4	Cards numbered 1 to 10. One is picked. **Event:** at least 7	_____	_____

5 A die is rolled. Are "rolling an even number" and "rolling a 1" complementary events? Explain.

Working

Yes / No because _____

EXAMPLE 2 — Fractions, decimals and percentages

Use me for **Round 2** ↓

a P(A) = $\frac{3}{8}$

$\frac{8}{8} - \frac{3}{8} = \boxed{\frac{5}{8}}$

b P(A) = 0.35

1.00 − 0.35 = **0.65**

Write 1 as 1.00 to line up.

c P(A) = 72%

100% − 72% = **28%**

1 whole = 100%

In each part, find P(not A). Simplify a fraction answer if you can, e.g. $\frac{9}{15} = \frac{3}{5}$.

ROUND 2 — Medium — Extension

↑ Use Example 2

6 P(A) = 0.23. Find P(not A).

P(not A) = 1.00 − 0.23 = _____

7 P(A) = $\frac{4}{9}$. Find P(not A).

Working

Answer: _____

8 The chance of rain is 64%. Find the chance of **no** rain.

Working

Answer: _____

9 P(the train is on time) = 0.875. Find P(the train is **not** on time).

Working

Answer: _____

10 P(win) = $\frac{6}{15}$. Find P(not win) in **simplest form**.

Working

Answer: _____

EXAMPLE 3 More than two outcomes

Use me for Round 3 ↓

A spinner is red, blue or green. P(red) = $\frac{1}{4}$ and P(blue) = $\frac{1}{3}$. Find P(green) and P(not green).



(1) All the outcomes add to 1. Use the same bottom number: $\frac{1}{4} + \frac{1}{3} = \frac{3}{12} + \frac{4}{12} = \frac{7}{12}$

(2) P(green) = $\frac{12}{12} - \frac{7}{12} = \boxed{\frac{5}{12}}$ P(not green) = $1 - \frac{5}{12} = \boxed{\frac{7}{12}}$ (red or blue)

ROUND 3 Challenging Extension

↑ Use Example 3

11 A spinner is red, blue or yellow. P(red) = 0.2 and P(blue) = 0.45.

0.2 + 0.45 = _____

P(yellow) = 1 − _____ = _____

P(not yellow) = _____

12 How students get to school:

Bus	Car	Bike	Walk
35%	40%	10%	?

Find P(walk) and P(**not** car).

Working

Answer: _____

13 Only A, B or C can happen. P(A) = $\frac{1}{6}$ and P(B) = $\frac{1}{3}$. Find P(C) and P(**not** B).

Working

Answer: _____

14 Jo says: "The chance my team wins is 0.6 and the chance it loses is 0.5." Draws can't happen. Explain why Jo must be wrong.

Working

Because _____

EXAMPLE 4 Work backwards to find how many

Use me for Round 4 ↓

A bag has 30 lollies. P(**not** red) = $\frac{3}{5}$. How many red lollies are in the bag?

(1) P(red) = $1 - \frac{3}{5} = \frac{2}{5}$

(2) $\frac{2}{5}$ of 30: 30 ÷ 5 = 6, 6 × 2 = 12. **12 red lollies** Check: 18 not red, $\frac{18}{30} = \frac{3}{5}$ ✓

ROUND 4 Challenging Extension

↑ Use Example 4

15 A bag has 40 marbles. P(**not** blue) = $\frac{3}{4}$. How many are blue?

P(blue) = $1 - \frac{3}{4} = \frac{\square}{\square}$ $\frac{\square}{\square}$ of 40 = _____ blue

16 A class has 28 students. P(a student is **not** left-handed) = $\frac{6}{7}$. How many are left-handed?

Working

Answer: _____

17 200 raffle tickets are sold. P(Ana does **not** win) = 0.95. How many tickets did Ana buy?

Working

Answer: _____

18 A spinner has 20 equal parts. P(**not** red) = 0.65. How many parts are red?

Working

Answer: _____

CHALLENGE Challenging Extension (stretch)

↑ Use all the examples

19 A box has only red, blue and green counters. P(**not** red) = $\frac{5}{8}$ and P(**not** blue) = $\frac{3}{4}$. Find P(green).

Working

Answer: _____

20 P(A) is 3 times as big as P(not A). Find P(A) and P(not A).

Hint: draw a bar split into 4 equal parts.

Working

Answer: _____

ANSWERS **1** rolling 2 or less; 1, 2 **2** not a prime; 1, 4, 6 **3** not a multiple of 3; 1, 2, 4, 5, 7, 8 **4** less than 7 (6 or less); 1 to 6 **5** No: 3 and 5 are in neither event **6** 0.77 **7** 5/9 **8** 36% **9** 0.125 **10** 9/15 = 3/5 **11** P(yellow) = 0.35; P(not yellow) = 0.65 **12** P(walk) = 15%; P(not car) = 60% **13** P(C) = 1/2 (3/6); P(not B) = 2/3 **14** 0.6 + 0.5 = 1.1, more than 1 **15** P(blue) = 1/4; 10 blue **16** 1/7 of 28 = 4 **17** P(win) = 0.05; 10 tickets **18** P(red) = 0.35; 7 parts **19** P(red) = 3/8, P(blue) = 2/8; P(green) = 3/8 **20** P(A) = 3/4; P(not A) = 1/4